

SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS AND SCIENCE, AMROLI
TYBCA SEM 6
602 DATA ANALYTICS USING PYTHON
MODEL PAPER 1

[1 HOUR]

[25 MARKS]

Q-1 Do as directed

[08]

(A) Answer the following MCQ (Any Four)

[04]

1. Which of the following is an example of ordinal data?
A) Blood group B) City name C) Student grade (A, B, C) D) Roll number
2. Which variable is used as input in regression?
A) Dependent variable B) Target variable C) Independent variable D) Output variable
3. Which ML technique is used for grouping similar data points?
A) Classification B) Regression C) Clustering D) Forecasting
4. In supervised learning, which option correctly matches **Dataset, Features, and Labels**?
A) Dataset = output only, Features = model accuracy, Labels = input values
B) Dataset = collection of data, Features = input variables, Labels = correct output/target
C) Dataset = training algorithm, Features = final prediction, Labels = error value
D) Dataset = test data only, Features = classes only, Labels = hidden data
5. The sutra Sesanyan is used to find:
A) The product B) The quotient C) The remainder D) The square root

(B) Answer the following in short (Any Four)

[04]

1. Explain the difference between Univariate, Bivariate, and Multivariate analysis.
2. Differentiate between dependent and independent variables in regression.
3. Explain overfitting and underfitting in Machine Learning.
4. Explain the term Labeled Dataset with example.
5. Explain why Chidana is considered a powerful Vedic Mathematics technique.

Q-2 (A) Discuss the importance of handling missing data and outliers in data analysis. Explain various methods used for data cleaning.

[05]

OR

Q-2 (A) Explain the Machine Learning life cycle in detail.

[05]

Q-2 (B) Explain Exploratory Data Analysis (EDA) and describe its objectives, importance, and steps involved in performing EDA.

[04]

Q-3 Answer the following (Any Two)

[08]

1. Describe quantitative and qualitative data. Explain their types and importance in data analytics.
2. Discuss the difference between covariance and correlation in statistical analysis. Explain with a practical dataset example.
3. In regression, what is **R² (Coefficient of Determination)**? Explain its meaning for **R² = 1, R² = 0, and negative R²** with an example.

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MODEL PAPER 2

[1 HOUR]

[25 MARKS]

Q-1 Do as directed

[08]

(A) Answer the following MCQ (Any Four)

[04]

1. Which type of EDA analyzes a single variable?
A) Univariate Analysis B) Bivariate Analysis C) Multivariate Analysis D) Regression Analysis
2. In regression, the variable being predicted is called:
A) Independent variable B) Dependent variable C) Control variable D) Random variable
3. Supervised learning requires:
A) Only input data B) Only output labels C) Both input data and output labels D) No labels at all
4. The sutra “Nikhilam Navatashcaramam Dashatah” is mainly used for:
A) Squaring numbers ending in 5 B) Multiplying numbers close to 10, 100, 1000 C) Finding square roots
D) Solving linear equations
5. Using Ekadhikena Purvena, what is 35^2 ?

(B) Answer the following in short (Any Four)

[04]

1. What is Qualitative Data? Explain Nominal and Ordinal data with examples.
2. Explain the role of training and validation data in machine learning.
3. What is the main difference between MSE and MAE?
4. How does the sutra Sesanyan help in divisibility and verification?
5. Explain the importance of Shunyam Saamyasamuccaye (शून्यं साम्यसमुच्चयै) in equation solving.

Q-2 (A) Discuss the different types of Exploratory Data Analysis with suitable examples.

[05]

OR

Q-2 (A) Explain the concept of data distribution. Discuss normal distribution, skewed distribution, skewness, and kurtosis.

[05]

Q-2 (B) Differentiate between supervised and unsupervised learning. Discuss their techniques, applications, and examples.

[04]

Q-3 Answer the following (Any Two)

[08]

1. Discuss the role of correlation in feature selection and predictive modeling. Explain positive, negative, and no correlation with examples
2. Write a note on the loss functions.
3. Differentiate between classification and regression